Tarun Mittal

Senior Software Engineer

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Summary

- Senior Software Engineer with **5 years** of experience designing and owning cloud-native, high-throughput **distributed systems**.
- Expertise in **Python and Go** microservices, with applied experience in **AI-assisted** and **RAG** systems.
- Proven track record shipping production systems, leading backend migrations, and improving reliability, performance, and scalability.

Skills

- **Languages** – Python, Go (Golang), SQL, C, C++.
- **Backend & APIs** – FastAPI, Gin, Django, REST, gRPC, Pydantic, Celery, asyncio.
- **Generative AI** – LangGraph, LangChain, RAG, Milvus, pgvector, LLM Agents (OpenAI/Gemini/Claude), Tool Calling, MCP, Prompt Engineering, LangSmith, Offline Evaluation.
- **Data & Streaming** – Apache Kafka, Apache Flink, ClickHouse, BigQuery, ETL Pipelines.
- **Databases & Storage** – PostgreSQL, MySQL, Redis, MongoDB, Elasticsearch, DynamoDB, S3.
- **Cloud & DevOps** – GCP, AWS, Docker, Kubernetes, Bazel, CI/CD, Datadog, ELK, New Relic.
- **Architecture & Core Concepts** – Distributed Systems, Microservices, System Design (HLD/LLD), Data Structures & Algorithms, Caching, Reliability, Multithreading, Database Sharding, Concurrency, Idempotency.

Professional Experience

Senior Software Engineer | IMPACT ANALYTICS, Bangalore | May 2026 – Present

AssortSmart — Senior Software Engineer (Platform & AI)

Platform Engineering & Infrastructure

- Architected a multi-tenant **Go/Gin** platform scaling to **10k peak RPS**, utilizing **Google Wire** for compile-time DI. Deployed as a self-protecting HTTP edge without a reverse proxy, guaranteeing high availability through native **h2c**, nested timeouts, and **Datadog** distributed tracing.
- Accelerated retail analytics performance by **15.5x**, slashing pivot query latency from **189s to 12s** on **250M-row** operations. Migrated the data layer to **ClickHouse**, scaling the catalog to process **1.6M** article-season combinations and **2.4B** weekly rollup rows.
- Eliminated deployment bottlenecks by **architecting a dynamic KPI configurator** and custom formula parser. Engineered a real-time compilation layer that tokenizes operator-authored math into safe, `ifNotFinite`-wrapped ClickHouse SQL fragments, **decoupling business logic from code releases**.
- Guaranteed strict **multi-tenant data isolation** and UAM-scoped hierarchy **access control** across **PostgreSQL** and **ClickHouse**. Secured system entry by implementing a Redis-fronted Firebase Admin to JWT/Google OIDC verification waterfall, constant-time API keys, and Postgres-backed role resolution.
- Enforced zero-regression engineering practices by designing a **100.0%** statement-coverage CI gate (race and atomic) across **1,200+** Go tests. Hardened the Bitbucket deployment pipeline with automated golangci-linting, pre-push hooks, and continuous SAST/SBOM security scanning.

Agentic Flows & Orchestration

- Architected a **multi-pipeline AI merchandise decision engine (Keep/Drop, Missed Opportunities, Top Style)** evaluating **335K+** products across multiple seasons. Optimized inference at scale—processing **88k** items per pass for under **$100** token spend—by blending deterministic KPI math with structured LLM invokes across **7 AI lenses**.
- Guaranteed zero-data-corruption across a **2.11B-row** ClickHouse fact table by physically air-gapping batch LLM execution from database queries. Ensured fault tolerance using ClickHouse `ReplacingMergeTree` for two-phase inserts, feeding models via JSON payloads and falling back to deterministic scores upon LLM failure.
- Engineered a **resilient AI orchestration** registry to safely manage high-throughput LLM workloads across millions of tokens. Guaranteed stable parallel batch processing via custom circuit breakers and durable checkpoints, embedding **LangSmith** and custom per-run JSON telemetry to track exact token costs, step durations, and batch fallbacks.
- **Shipped "Ask Iris," a WebSocket AI copilot** enabling planners to interactively query multi-billion-row KPIs, generate dynamic charts, and drill down into AI decisions. Architected the underlying **LangGraph** orchestration—featuring a Supervisor router and Evaluator loop—enforcing socket-level frozen scopes to guarantee data isolation and prevent infinite LLM loops.
- Established rigorous **AI deployment guardrails** by building a **300-case** offline evaluation harness and a **≥80%** accuracy CI promotion gate. Prevented degraded rollouts by benching candidate agent configurations against a **74%** deterministic baseline, ensuring provable business value before production release.
- **Tech** – Go, Gin, Python, FastAPI, LangGraph, ClickHouse, BigQuery, Redis, PostgreSQL, Datadog, LangSmith, GCP, Docker.

Software Development Engineer 2 | UBER (via EPAM Systems), Bangalore | July 2024 – May 2026

Financial Risk Management (FRM) Scoping Platform

- Owned the Financial Risk Management scoping backend, reducing quarterly reconciliation time by **70%** (from **14 days to 3 days**) against a **$340M** materiality threshold. Engineered the Sheets-to-MySQL migration across **19M** raw GL rows, shipping **36 FastAPI** endpoints to automate custom financial math and a nested 4-level FSLI (L1–L4) hierarchy with dynamic quarter-annualization.
- Resolved ETL pipeline lock contention during financial close by designing an 8-table normalized MySQL schema on Uber's SOADB featuring a polymorphic review-status model. Guaranteed real-time, multi-user collaboration integrity by implementing optimistic row-locking, atomic cross-table syncs, and deterministic SHA-256 natural-key hashing to eliminate data duplication across quarterly audit trails.
- Led **3 engineers** in a core SQLAlchemy 2.0 ORM migration that eliminated latent column-aliasing bugs and raised module statement coverage to **100%**. Orchestrated zero-downtime releases via stacked Bazel PRs, embedding reporting SQL allowlists and automated FSM gates with **50%** delta-variance checks to enforce SOX compliance.
- **Tech** – Python, FastAPI, SQLAlchemy 2.0, Pydantic, MySQL, Bazel, Docker.

Menu Ingestion & Automation (Uber Eats)

- Accelerated restaurant onboarding from **24 hours to 2 hours** and saved **$600K** annually by architecting an automated ingestion pipeline processing **30K+** menus monthly. Converted unstructured, multilingual PDFs into a strict catalog schema at **98%** field fidelity utilizing a **Gemini 2.5 Pro** and **LangChain RAG** pipeline over **Milvus** vector embeddings.
- Scaled menu extraction success to **95%+** by hardening a Selenium fleet with dynamic proxy pools and adaptive retries. Designed a **large-scale data ingestion and processing pipeline** using **Kafka** and **Apache Flink**, with stateful keyed deduplication and schema normalization to guarantee exactly-once catalog upserts.
- **Tech** – Python, Selenium, Kafka, Flink, LangChain, Gemini, RAG, Milvus, GCP, Docker.

Document Compliance (Uber Mobility)

- Achieved **99.9%** regulatory compliance for Uber Mobility ANZ by automating driver and vehicle document verification workflows against local authority requirements. Eliminated **20 hours** of manual operational overhead per week by replacing manual review queues with a deterministic validation backend.

Software Development Engineer 2 | MASTERS INDIA, Noida | December 2022 – June 2024

GST Compliance & E-Invoicing Platform

- Reduced p95 API latency by **75%** (from **1.2s to 300ms**) across **1,500+** enterprise clients by spearheading the migration of a legacy PHP monolith to a **FastAPI** microservices architecture. Mentored two engineers through the transition, optimizing backend design to drive overall system throughput from **700 to 4,000 requests per minute**.
- Architected a **high-concurrency** bulk e-invoicing pipeline processing **1M+** daily government submissions by orchestrating a **Kafka** event bus and a **PostgreSQL** database horizontally sharded by tax quarter. Enforced exactly-once, **idempotent** processing for **100K+** concurrent imports with **fault-tolerant state management** via dead-letter queues and bounded retries against volatile state IRP portals.
- Decreased incident triage time by **70%** and lowered database read load by **30%** by instrumenting **ELK** and **New Relic** distributed tracing alongside a Redis caching layer. Hardened the CI/CD release cycle by driving test coverage from **35% to 82%**, achieving a **98%** deployment success rate across the microservices ecosystem.
- **Tech** – Python, FastAPI, Kafka, PostgreSQL, MongoDB, Redis, Elasticsearch, Celery, Docker, ELK, New Relic, AWS.

Software Development Engineer | GEEKSFORGEEKS, Noida | August 2021 – November 2022

Courses Platform and Influencer Dashboard

- Stabilized the doubt-support platform to handle **10K+** daily queries and **10x** traffic spikes during live coding contests by migrating the backend to **Django**. Engineered high-throughput REST APIs for voting, pinning, and locking threads utilizing MySQL, MongoDB, Redis, and Elasticsearch, driving a **20%** lift in premium subscriptions.
- Accelerated platform course sales by **30%** by launching an influencer analytics and earnings dashboard to dynamically track transactions and affiliate coupon conversions. Increased engineering operational efficiency by **70%** by designing asynchronous scheduled workflows for automated video processing, user reminders, and data cleanup.
- **Tech** – Python, Django, MySQL, MongoDB, Redis, Elasticsearch.

Education

B.Tech (Information Technology) | **Institute of Engineering & Technology, Lucknow** | **July 2017 – June 2021**

Achievements & Certifications

- Ranked **2260 / 37,000+** in the Google Code Jam Qualifier 2021 and finalist (top 3 nationally) at the **Smart India Hackathon 2020 (MHRD)**.
- Participated in the **Global AI Hackathon** at EPAM Systems.
- Earned **HackerRank Problem Solving** and **LangChain Academy** certifications (HackerRank, LangChain).